

STEP 1 OF 3 · FOR APPROVAL

How to Write a Perfect Medical Prompt

The prompt-writing guidelines behind the 1,000-prompt library for doctors, residents and medical students, built from the published prompting guidance of the four major AI labs.

OpenAI · ChatGPT

Anthropic · Claude

Google · Gemini

Microsoft · Copilot

11

principles all four labs
agree on

6

building blocks in every
prompt (R-C-T-S-F-Q)

1,000

prompts across 3 parts and
27 sections

CONTENTS

What's inside

- 01 What the four AI labs agree on
- 02 Each lab's framework in one line
- 03 The master structure: R-C-T-S-F-Q
- 04 Length and context rules: how many words, how much context
- 05 Format rules: how each prompt is written
- 06 Medical safety and accuracy rules
- 07 The prompt card template
- 08 Worked examples: one per library part
- 09 Quality checklist
- 10 The 1,000-prompt library plan
- 11 Sources

How to read this document. Sections 1–2 summarise the research. Sections 3–9 are the rules I will follow when writing every prompt. Section 10 is the proposed structure of the final 1,000-prompt PDF, which I'd like you to approve or change before I start.

What happens next. ① You approve these guidelines. ② I study the prompt PDFs already in your Google Drive and send a combined "how I'll write the prompts" PDF. ③ I write the 1,000 prompts.

What the four AI labs agree on

I read the official prompting guides from OpenAI, Anthropic, Google and Microsoft. Their vocabulary differs, but they agree on the core, and any prompt that follows these principles works well in any chatbot.

#	Principle	OpenAI	Anthropic	Google	Microsoft
1	Be clear, specific and direct. Vague prompts get generic answers.	✓	✓	✓	✓
2	Give the model a role or persona.	✓	✓	✓	✓
3	Give context: who you are, the purpose, the audience and the constraints.	✓	✓	✓	✓
4	State the output format: structure, length and tone.	✓	✓	✓	✓
5	Separate the sections with delimiters (### or XML tags).	✓	✓	✓	✓
6	Use examples (few-shot) when format or style matters.	✓	✓	✓	✓
7	Break complex tasks into ordered steps or a chain of prompts.	✓	✓	✓	✓
8	Say what to do, not only what to avoid.	✓	✓	✓	✓
9	Ground the answer in the source material and name the source.	✓	✓	✓	✓
10	Iterate: test, refine, and use follow-up turns.	✓	✓	✓	✓
11	Avoid contradictory instructions. New models follow instructions literally.	✓	✓	-	-

Each lab's framework in one line

GOOGLE GEMINI

PTCF: Persona · Task · Context · Format

Google reports that successful Workspace prompts average **about 21 words**, and that complex requests need longer, more detailed prompts. Its advice is to write in natural language, as if talking to a person.

MICROSOFT COPILOT

GCES: Goal · Context · Expectations · Source

A goal alone is enough to get an answer. Context, expectations (format, tone, length) and a named source make the answer more specific and better grounded.

OPENAI CHATGPT

Identity → Instructions → Examples → Context

Put instructions first and separate the sections with ### or XML. GPT-5 follows instructions "with surgical precision", so contradictions hurt more than they used to. Length and verbosity can be set in plain language.

ANTHROPIC CLAUDE

Clear & direct · explain why · XML · examples

Treat the AI as "a brilliant but new employee who lacks context". Explain the *why* behind each instruction, give 3–5 diverse examples, put long documents first and the question last (**up to 30% better**), and ask it to quote the source before answering.

MEDICAL LITERATURE

JMIR 2025 clinician tutorial · medical-education papers

These describe the same four pillars (Role, Context, Task, Output), add "think step by step" for clinical reasoning, and warn about **phantom citations**: every AI-generated reference has to be checked in PubMed or another verified database.

03 · THE MASTER STRUCTURE

R-C-T-S-F-Q: six blocks in every prompt

To follow all four frameworks at once, every prompt in the library will use the same six labelled blocks:

R
Role
Who the AI should be

C
Context
Who I am, my situation, why I need this

T
Task
The exact action, as numbered steps

S
Specifics
Guidelines, constraints, sources

F
Format
Structure, length, tone

Q
Quality
Accuracy guard + next step

How it maps to each lab's framework

R-C-T-S-F-Q	Google PTCF	Microsoft GCES	OpenAI	Anthropic
Role	Persona	-	Identity	<role>
Context	Context	Context + Goal	Context	<context>
Task	Task	Goal	Instructions	<task> / <instructions>
Specifics	Context	Source	Instructions	<constraints>
Format	Format	Expectations	Output format	<output_format>
Quality	-	-	Self-check	"Quote first", "say if unsure"

04 · LENGTH & CONTEXT

How many words, and how much context?

None of the labs sets a single ideal word count. Google's ~21-word average describes quick everyday requests, and all four labs say complex tasks need more detail. The library therefore uses **three tiers**:

Quick
30-60 words
One focused output: a mnemonic, 10 MCQs, one differential. Role + Task + Format in 2-4 sentences.

Standard · default
80-150 words
Most prompts: a thesis section, a seminar outline, a study plan. All six blocks as short labelled lines.

Deep
150-300 words
Multi-step work: a full ROL framework, a protocol, a mock viva. Numbered steps and an optional example.

Five rules for context

1. **Enough to remove guesswork, and no more.** Give 2–5 facts the AI can't infer: your level (MBBS / PG year / consultant), specialty, topic, deadline or exam, audience, and the standard you need followed (ICMJE, CONSORT, STROBE, PRISMA, CARE, Vancouver).
2. **Explain the why.** "...because this is for my MD thesis committee" gives a better result than "be formal". (Anthropic)
3. **Fill-in blanks go in [SQUARE BRACKETS]**, e.g. [SPECIALTY], [TOPIC], [STUDY DESIGN], [EXAM]. Keep it to 5 or fewer per prompt.
4. **Pasted material goes first and the instruction goes last.** When the user pastes a paper or data, the paste slot sits above the final instruction. (Anthropic long-context rule)
5. **No contradictions.** Never "be concise" and "be exhaustive" in the same prompt. (OpenAI GPT-5 guide)

05 · FORMAT RULES

How each prompt is written

Start with an action verb

Draft, Build, Critique, Convert, Simulate, Compare, Tabulate. It tells the AI exactly what kind of work to do.

Phrase instructions positively

"Write in formal academic prose" rather than "Don't be casual". All four labs recommend this.

Number the steps

Whenever order or completeness matters, list the steps 1), 2), 3) so none get skipped.

Name the output shape

"A table with columns X | Y | Z", "8–10 slides with speaker notes", "300 words", "5 viva questions with model answers".

Plain labelled lines

Role: / Context: / Task: ... These paste cleanly into any chat app and every model parses them reliably. XML is optional for advanced users.

One prompt, one job

A big job becomes a chain (Prompt 1 → 2 → 3) rather than one giant prompt, as both Anthropic and OpenAI recommend.

Interaction hook. Where it helps, prompts end with "Ask me 3 clarifying questions before you start" or "End by suggesting the next step", so the chat keeps improving instead of stopping at a first draft.

Mandatory guards for medical prompts

Every research or clinical prompt includes at least one of these:

Guard	Wording used in prompts
Citation guard	"Do not invent references. Only cite papers you are confident exist, give the DOI/PMID, and mark anything uncertain as [VERIFY]." Users are told to check every reference in PubMed.
Uncertainty permission	"If you are unsure or the evidence conflicts, say so explicitly."
Guideline anchoring	"As per the latest [ACC/AHA / NICE / WHO / ICMR] guideline..."
Reasoning request	"Reason step by step, showing how each finding changes the probability of each diagnosis."
Privacy	"Use de-identified details only." No patient names, IDs or dates.
Academic integrity	Thesis and manuscript prompts help users structure, critique and improve their own work. They never ask the AI to fabricate data or results, and they remind users to follow their institution's and journal's AI-disclosure policy.
Clinical disclaimer	Diagnostic prompts are for learning and decision support, not a substitute for clinical judgement.

The prompt card: how every prompt will look

#[NUMBER]	[SHORT TITLE]	Tier: Quick Standard Deep
Section >	Subsection	Best for: Student Resident Faculty
Role:	You are a [expert persona with relevant credentials].	
Context:	I am a [level] in [SPECIALTY] working on [TOPIC/GOAL] for [PURPOSE/AUDIENCE].	
Task:	[Action verb] ... 1) ... 2) ... 3) ...	
Specifics:	Follow [GUIDELINE/STANDARD]. [Constraints]. [Accuracy guard].	
Format:	[Structure], [length], [tone].	
Next:	[Clarifying-question hook or follow-up suggestion]	
Tip:	[one-line pro tip on how to use or chain this prompt]	

One example from each part of the library

A · Thesis & Research: methodology chapter (Standard)

Role: You are a senior medical research methodologist and thesis guide.

Context: I am a PG resident in [SPECIALTY]. My thesis topic is [TOPIC], a [STUDY DESIGN] with [SAMPLE SIZE] patients. My committee wants a clear, well-justified methodology chapter.

Task: Draft the methodology outline: 1) study design and setting, 2) inclusion/exclusion criteria, 3) sample size justification, 4) variables and definitions, 5) data collection, 6) statistical plan, 7) ethics.

Specifics: Align with the [STROBE/CONSORT] checklist. Flag any gap in my design. Do not invent references; mark uncertain ones [VERIFY].

Format: Numbered headings with 2–4 bullet points each, formal academic tone, ~400 words.

Next: Ask me for any missing details before finalising.

B · Seminars & Journal Clubs: journal club presentation (Standard)

Role: You are an experienced journal-club moderator and critical-appraisal expert.

Context: I am presenting the paper below at our [SPECIALTY] department journal club to residents and faculty.

[PASTE ABSTRACT / FULL TEXT HERE]

Task: 1) Summarise the PICO, 2) appraise it using the [CASP/JBI] checklist, 3) list 3 strengths and 3 limitations, 4) state the clinical bottom line, 5) suggest 5 discussion questions.

Format: 12–15 slide outline with slide titles, bullet content and one-line speaker notes.

Next: End with the 3 questions faculty are most likely to ask me, with short answers.

C · Study & Exam Prep: mock viva (Deep)

Role: You are a strict but fair [SPECIALTY] examiner for the [EXAM] viva.

Context: I am a final-year PG preparing for my viva in [WEEKS] weeks. My weak area is [TOPIC].

Task: Conduct a mock viva. Ask one question at a time, starting basic and escalating to examiner-level. After each of my answers: 1) score it /10, 2) give the ideal answer in 3–5 points, 3) name the gap in my reasoning, then ask the next question.

Specifics: Base standards on current guidelines ([GUIDELINE]). If a fact is guideline-dependent or debated, say so.

Format: Conversational. Keep each question under 40 words.

Next: After 10 questions, summarise my weak areas and give a 3-day revision plan.

Checklist every prompt must pass

- ☐ Has a Role, Context, Task and Format (the four elements every lab agrees on)
- ☐ Task starts with an action verb and is clear enough that a colleague could follow it (Anthropic's "golden rule")
- ☐ Falls in the right word tier: Quick 30-60 / Standard 80-150 / Deep 150-300
- ☐ Placeholders are in [BRACKETS], 5 or fewer
- ☐ No contradictory instructions
- ☐ Output shape is named (table, slides, word count, number of items)
- ☐ Includes a medical accuracy guard where relevant (citations, guidelines, uncertainty)
- ☐ Works unchanged in ChatGPT, Claude, Gemini and Copilot
- ☐ Doesn't duplicate another prompt in the library
- ☐ Positive phrasing: says what to do

The 1,000-prompt library: proposed structure

#	Section	Prompts
1	Thesis, Research & Publication	400
1.1	Topic selection & research question (PICO / FINER, gap finding)	35
1.2	Literature search strategy (PubMed / MeSH, Boolean, databases)	30
1.3	Writing the Introduction	30
1.4	Review of Literature (ROL)	40
1.5	Aims, objectives & hypotheses	20
1.6	Study design & methodology	40
1.7	Sample size & statistical analysis	45
1.8	Results, tables & figures	25
1.9	Discussion, conclusion & limitations	30
1.10	Protocol, ethics & synopsis	20
1.11	Thesis to manuscript & journal publication	35
1.12	Case reports & case series	30
1.13	Conference abstracts, posters & oral papers	20
2	Seminars, Journal Clubs & Presentations	300
2.1	Seminar preparation (any topic, any specialty)	60
2.2	Journal club: selection, critical appraisal, presentation	60
2.3	Slide design & PowerPoint structure	40
2.4	Conference oral presentations	35
2.5	Conference posters (e-posters & print)	35
2.6	Case presentations & grand rounds	40
2.7	Delivery, Q&A handling & teaching sessions	30
3	Study, Exam & Viva Preparation	300
3.1	Learning any topic (concept mastery, active recall)	50
3.2	Study planning & revision schedules	30
3.3	MCQ practice & exam strategy (NEET-PG, INI-CET, USMLE, PLAB, MRCP...)	50
3.4	Long / short answer & essay writing	30
3.5	Viva & OSCE preparation	50
3.6	Clinical reasoning & differential diagnosis	60
3.7	Memory aids, flashcards & high-yield summaries	30
	Total	1,000

Sources

- **Anthropic**, Prompting best practices (Claude Platform Docs): <https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/claude-prompting-best-practices>
- **Anthropic**, Prompt engineering best practices for 2026: <https://claude.com/blog/best-practices-for-prompt-engineering>
- **OpenAI**, GPT-5 prompting guide (OpenAI Cookbook): https://github.com/openai/openai-cookbook/blob/main/example/gpt-5/gpt-5_prompting_guide.ipynb
- **OpenAI**, Prompt engineering guide: <https://developers.openai.com/api/docs/guides/prompt-engineering>
- **OpenAI Help Center**, Best practices for prompt engineering with the OpenAI API: <https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api>
- **Google**, Prompt design strategies (Gemini API): <https://ai.google.dev/gemini-api/docs/prompting-strategies>
- **Google**, 5 ways to write better AI prompts for Gemini in Workspace: <https://blog.google/products-and-platforms/products/workspace/google-gemini-workspace-ai-prompt-tips/>
- **Microsoft**, Get started writing prompts in Microsoft 365 Copilot: <https://support.microsoft.com/en-us/microsoft-365-copilot/get-started-writing-prompts-in-microsoft-365-copilot>
- **Microsoft**, Prompt engineering techniques (Azure OpenAI / Foundry): <https://learn.microsoft.com/en-us/azure/foundry/openai/concepts/prompt-engineering>
- **JMIR (2025)**, Prompt Engineering in Clinical Practice: Tutorial for Clinicians: <https://www.jmir.org/2025/1/e72644/PDF>
- **PubMed**, Prompt engineering in medical education: digestive surgery: <https://pubmed.ncbi.nlm.nih.gov/42417731/>

Companion file. The same rules are packaged as **SKILL.md** (medical-prompt-writer), so every one of the 1,000 prompts is generated and checked against the same standard.